E^2-JEPA: From Free Energy to Automatic Hierarchy

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Abstract

Long-horizon control requires hierarchy, and hierarchy requires knowing which latent dimensions stay valid for long horizons. We measure this directly—the relaxation time τ of each latent dimension—and find a universal failure: across five self-trained JEPA regimes and an independent published world model, the sorted τ spectrum never shows a gap larger than 1.03, so no hierarchy boundary can be placed anywhere. The diagnosis is causal: the underlying dynamics *do* contain multi-scale structure, but a scalar-temperature entropy regularizer cannot organize the coordinates, because compressed representations leave no room for organization while overcomplete ones are homogenized by dense normalization. Entropy is moreover an existence precondition: collapsed representations exhibit pseudo-slow τ with zero information, so any fix must live inside a free-energy objective and act differentially across scales. We therefore constructed the first-principles candidate under these constraints—a **temperature vector**, $F = U - \sum_k T_k S_k$ over latent bands—and tested it under preregistered acceptance criteria. It fails informatively: every redistribution metric moves in the intended direction, but the inter-band ratio reaches only 1.11 against a required 1.5, because temperature governs the *survival* of information, not its *placement*, while the prediction energy can only remove content, never place it. Organizing latents by time scale therefore remains genuinely open; this proposal contributes the referee (the τ protocol), the discipline (preregistered falsification loops), and the endgame (a subgoal backtest) with which to explore it.

1. Introduction

Joint-Embedding Predictive Architectures (JEPAs) learn world models by predicting in representation space rather than observation space, avoiding the cost of pixel-level reconstruction while producing compact latents suitable for planning [4, 1]. For long-horizon control, however, prediction alone is not enough: the planner must decide *at which temporal scale each piece of latent information is valid*. When hierarchical world models ignore this and treat upper-level predictions directly as subgoals for the lower level, policy success degrades from 73.3% (oracle subgoals) to 38.7% [7]—the interface fails precisely because nothing marks which latents are coarse enough to serve as goals and which are not.

Recent work has begun interrogating the forward-prediction objective itself. PSG-JEPA observes that forward prediction does not enforce reliable identifiability of physical state, and adds proprioceptive grounding as a training-time auxiliary [6]. Our work addresses the orthogonal deficiency that remains even when latents are identifiable: *temporal unorganization*. We ask: *can the training objective itself organize latent dimensions by their characteristic time scale, so that hierarchy emerges from first principles rather than architectural heuristics?*

2. A Measurable Notion of Temporal Scale

We define the granularity of a latent dimension as its characteristic relaxation time τ_i: the duration over which dimension i stays correlated with its own past. This is the physicist's correlation time—the quantity that separates slow from fast modes in statistical mechanics and sets the coarse-graining axis of the renormalization group—lifted from molecular observables to latent coordinates. We estimate τ with two independent estimators, because each is blind to what the other sees.

Estimator A (data side). For each dimension, integrate its autocorrelation function $C_i(\Delta)$ over lags Δ, truncating at the first non-positive lag (the initial-positive-sequence rule, which prevents noise at large lags from cancelling real signal). This estimator uses trajectories only; no model is involved.

Estimator B (model side). Linearize the learned one-step latent dynamics f at data points to obtain the Jacobian $J = \partial f/\partial z$, and convert each eigenvalue to a time scale via $\tau = -1/\ln|\lambda|$. This estimator uses the model only; no data statistics are involved.

Why the metric can be trusted. Three properties, established in Section 3, carry it. First, the two estimators are independent—one reads data trajectories, the other reads the learned dynamics—so agreement certifies structure and disagreement localizes artifacts. Second, the protocol calibrates against ground truth where one exists: on the external model it recovers exactly the dimension of the physical state (Section 3.2), a number known independently of anything we trained. Third, the metric declares its own failure modes: it is undefined on collapsed representations rather than silently returning pseudo-slow readings (Section 3.4). A measurement that can be calibrated, cross-checked, and made to fail loudly is one whose positives can be believed. A hierarchy boundary, if any, should be placed at the largest multiplicative *gap* in the sorted τ spectrum; the empirical finding is that this gap does not exist. Section 4 reports the first attempt to create it, and what its failure teaches.

3. Preliminary Results: The Collapse Is Real, Universal, and Diagnosed

The measurement phase is complete. All numbers below are measured on PushT teleoperation data with the protocol of Section 2; none are projected.

3.1 Universal spectral gap collapse

We trained five regimes—energy-only (no entropy term), JEPA, E^2-JEPA, behavior cloning (BC), and flow-matching BC—under identical data. We additionally audited the published LeWM PushT world model [5], which is deployed for model-predictive control, on its own released training set. In every case the sorted τ spectrum is gapless: the maximum ratio between consecutive values is ≤ 1.03 (Fig. 1). No training objective, no model scale, and no anti-collapse mechanism produced a natural hierarchy boundary.

3.2 The dynamics know the scales; the coordinates do not inherit them

On the same E^2-JEPA latents, estimator A finds every dimension within $\tau = 9 \pm 1$ environment steps, while estimator B finds a spectrum spanning 0.1–20 environment steps (Fig. 2). The mismatch is the finding: the learned dynamics contain multi-scale structure, but the coordinate system does not express it. The external model sharpens this into a calibration result: LeWM's Jacobian spectrum contains exactly 7 neutral modes ($|\lambda| \geq 1$ at every sampled context), matching the physical state dimension of PushT (agent position, block pose, velocities)—the model learned

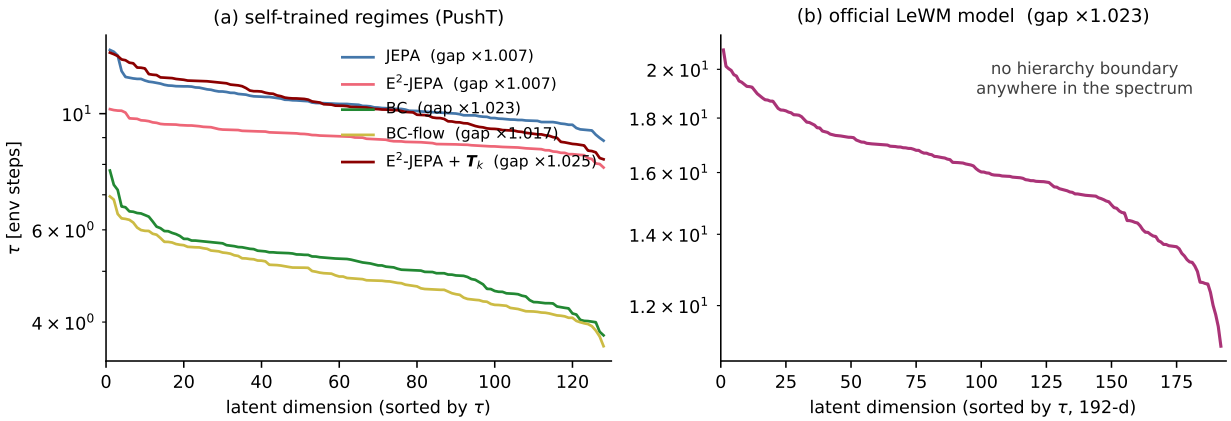


Figure 1: **Universal spectral gap collapse (measured).** (a) Sorted τ spectra of five self-trained regimes on PushT—the four preliminary arms plus the temperature-vector arm tested in Section 4; each annotation gives the largest consecutive ratio. (b) The official LeWM world model, measured on its own released training data under the identical protocol. No spectrum contains a boundary at which a hierarchy could be cut.

a unit-gain pass-through of the physical manifold—while its coordinate-level τ spectrum remains gapless. Structure exists in the dynamics; the objective never forces coordinates to inherit it.

3.3 Why scalar anti-collapse pressure cannot organize

Existing regularizers prevent *dimensional* collapse but are indifferent to temporal organization [2, 3]. Our measurements explain why this indifference is structural, through two complementary failure regimes (Fig. 3):

- **Compressed representations have no room to organize.** LeWM's latent has participation ratio $6.2 \approx$ the physical state dimension: the data manifold is so low-dimensional that any half of any basis spans the entire state. Measured consequence: decoding block pose from the τ-slow half vs. the τ-fast half gives $R^2 = 0.61$ vs. 0.60—flat, because a sufficient statistic cannot be split.
- **Overcomplete representations are homogenized.** Our E^2-JEPA latent has participation ratio 114, leaving room for organization, but dense normalization mixes every dimension with every other, collapsing all data-side τ to 9 ± 1.

Either too little room or too much mixing—both regimes block spontaneous organization, so organization must be imposed by the objective function. This is the necessity argument for what follows.

3.4 Entropy is an existence precondition for τ

Our energy-only control (free energy with the entropy term removed) collapses toward constant latents and measures *pseudo-slow* τ: median 15.6, larger than any healthy model, while carrying near-zero information (probe $R^2 \leq 0.42$, per-dimension std 0.003; Fig. 4). A constant signal has perfect autocorrelation and zero content. Hence τ is defined only where entropy is positive: the entropy term is not a regularization trick but the *existence precondition of the metric this proposal*

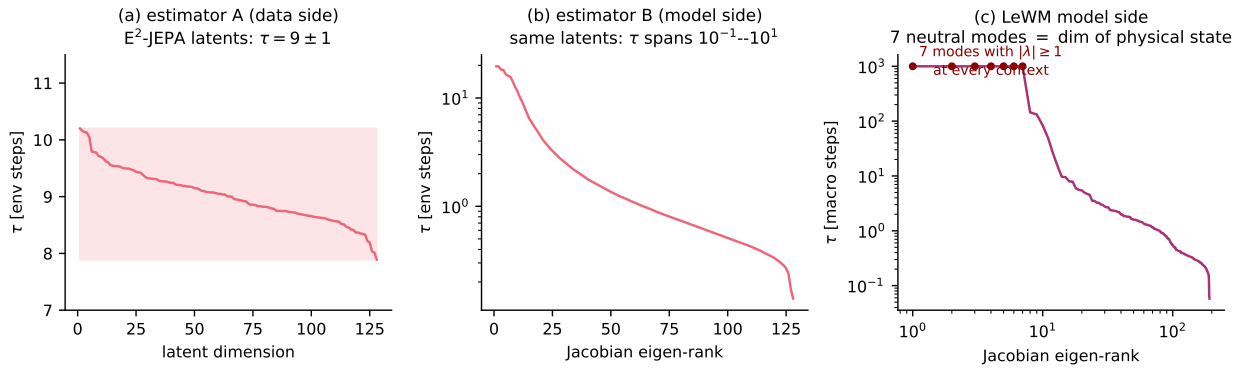


Figure 2: **A/B dissociation (measured).** (a) Data-side τ of E^2-JEPA latents: one narrow band. (b) Model-side τ of the same latents: three orders of magnitude. (c) LeWM's model-side spectrum: exactly 7 neutral modes, equal to the dimension of the physical state—the protocol recovers a known ground truth, and the dynamics provably contain the multi-scale structure that the coordinates lack.

is built on. Any mechanism that manufactures a τ spectrum must therefore live inside a free-energy objective—which fixes the design space for the method.

4. A First Mechanism, Tested: The Temperature Vector Is Insufficient

The preliminary results constrain any fix: it must (i) live inside a free-energy objective (Section 3.4), (ii) act differentially across time scales, because a uniform scalar cannot (Section 3.3), and (iii) exploit the structure already present in the dynamics (Section 3.2). The temperature vector is the natural first candidate under these constraints. A scalar T applies one entropy pressure to all dimensions; because total entropy is identical for a temporally mixed and a temporally separated representation, the scalar objective cannot distinguish them—it is invariant under any relabeling of dimensions, so no gradient direction points from "mixed" toward "organized." Replacing the scalar with a vector $\boldsymbol{T} = (T_1, \ldots, T_K)$ over K bands, each with its own per-dimension band entropy $S_k = \frac{1}{2|B_k|} \log\det \Sigma_k$,

$$F = U - \sum_{k=1}^{K} T_k \, S_k, \tag{1}$$

breaks that permutation symmetry and creates a differential pressure under which, in principle, a mixed spectrum is no longer stationary. We trained this arm and scored it against criteria fixed before training.

Preregistered test. K=4 contiguous bands of 32 dimensions; a geometric temperature gradient 8:4:2:1, normalized so the total entropy budget matches the scalar arm (only its distribution across bands differs); identical cooling schedule, data, batch, seed, and training protocol as the scalar E^2-JEPA arm. Criteria, fixed before the run: (1) inter-band ratio of band-median $\tau \geq 1.5$ with monotonically decreasing medians; (2) block-pose probe R^2 from the slow band exceeds the fast band by ≥ 0.2; (3) agent-position probe R^2 from the fast band ≥ 0.3. A collapse guard requires every band's mean std ≥ 0.01, so a degenerate solution cannot masquerade as success.

Result: all three criteria fail (Table 1), and the failure is clean. Every redistribution metric moved in the intended direction—band-median τ decreases monotonically $11.5 \to 9.2$, band

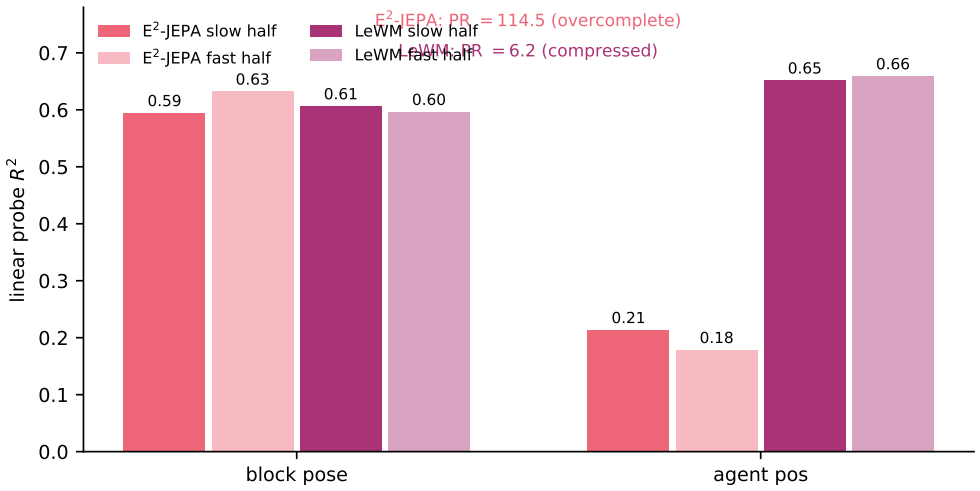


Figure 3: **Two dead ends (measured).** Linear-probe R^2 for block pose and agent position, from the τ-slow vs. τ-fast half of each spectrum. LeWM (compressed, PR = 6.2) cannot be split because any half spans the whole physical state; E^2-JEPA (overcomplete, PR = 114.5) has room but the halves are interchangeable because per-dimension τ is homogenized. Information is present in both; temporal organization is present in neither.

preregistered criterion	threshold	measured
inter-band ratio (monotone)	≥ 1.5	1.05 / 1.09 / 1.11
slow-band minus fast-band block probe	≥ 0.2	0.08
fast-band agent probe	≥ 0.3	0.10

Table 1: The temperature-vector arm, scored against its preregistered criteria under the scalar arm's protocol. All three fail; the collapse guard passes (band stds 2.14–0.83), so the negative is clean.

std decreases 2.14 $\rightarrow$ 0.83, the block probe declines across bands 0.59 $\rightarrow$ 0.46, the τ–block correlation strengthens to +0.29 (scalar arm: +0.07), and the participation ratio drops from 114.5 to 45.7, i.e. the representation is *less* homogenized—yet the spectra of the two arms overlap almost exactly (Fig. 5) and the fast variable is not retained anywhere.

Diagnosis: temperature governs survival, not placement. Entropy pressure decides whether information in a band survives the energy's incentive to drop it; it does not decide *which band* content belongs to. The energy, meanwhile, prefers slow content everywhere (it is predictable) and removes fast content everywhere (it is not): it can only remove, never place. A band protected enough to retain fast content therefore also retains slow content—the two-dead-ends argument of Section 3.3 reproduces *inside* the mechanism. The failure is thus not an implementation accident but the theory correctly predicting its own difficulty: organizing latents by time scale requires a force that *places* information, and no member of the temperature family provides one.

5. The Open Question, and How to Explore It

The failure sharpens the problem: what force can *place* information by time scale inside a trainable objective? This is the open core of temporal abstraction—if it were easy, hierarchical world models would not still treat their upper level's predictions as subgoals and fail [7]. We structure the

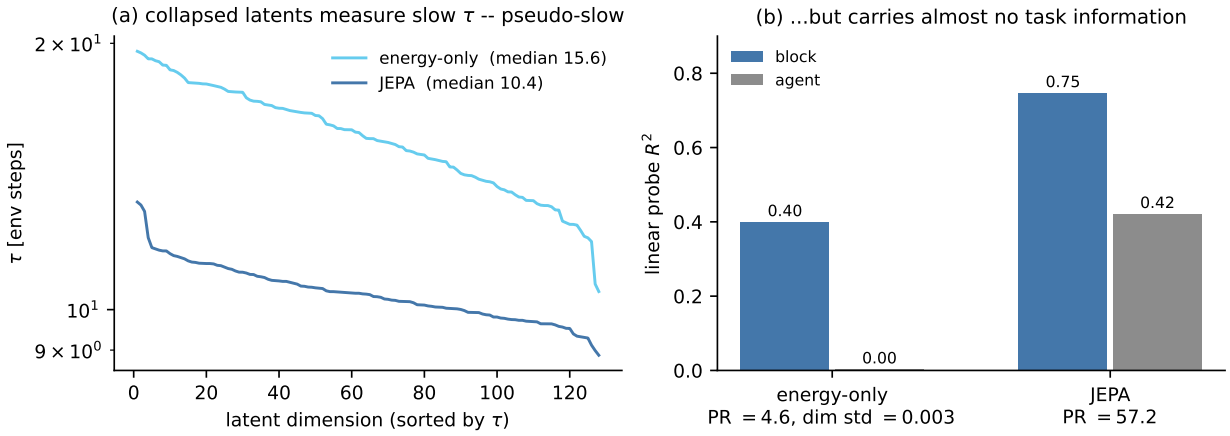


Figure 4: **Pseudo-slow τ (measured).** (a) Removing the entropy term yields latents that measure *slower* than any healthy model. (b) Yet the same latents decode almost nothing, and their variance is degenerate. τ without entropy is meaningless, so the free-energy form is not an analogy but the existence precondition of the metric.

exploration as preregistered falsification loops around three falsifiable questions:

1. **Open-loop or closed-loop fields.** The tested gradient was fixed a priori over fixed bands. Does coupling the temperature field to the measured spectrum—re-assigning bands by τ rank during training—escape the placement failure, or is every differential *survival* pressure subject to it?

2. **Coordinate side or dynamics side.** Fig. 2 shows the multiscale structure lives in the learned dynamics but is not inherited by the coordinates. Is enforcement more natural where the structure already is—on the eigen-structure of the one-step Jacobian—than on the coordinates it fails to reach?

3. **Objective or architecture.** If no objective-side force suffices, temporal organization is an architectural constraint, as hierarchy-by-design lines assume implicitly. The τ protocol can decide this empirically instead of by taste.

Each loop is a single-arm training plus one τ measurement, every acceptance criterion is fixed before the run, and the referee is the same protocol that audited the external model. When any mechanism passes, the plan converges on the endgame: hierarchy boundaries cut at the measured gap, and subgoals admitted only where their τ matches the lower level's reachable horizon, back-tested on the failing interface of [7] (38.7% vs. 73.3%)—the direct test of whether the metric repairs the interface it was motivated by. Transfer to a different embodiment is optional scope, not a claim.

6. Risk Assessment

The proposal has already absorbed its first preregistered falsification: the temperature vector failed, was diagnosed, and narrowed the hypothesis space—survival is not placement. The same discipline governs the remaining risk, that no objective-side force organizes latents at all. That terminal negative would itself be a result: it would locate temporal organization on the architectural side of the design space, and it would be published together with the protocol that proved it. The

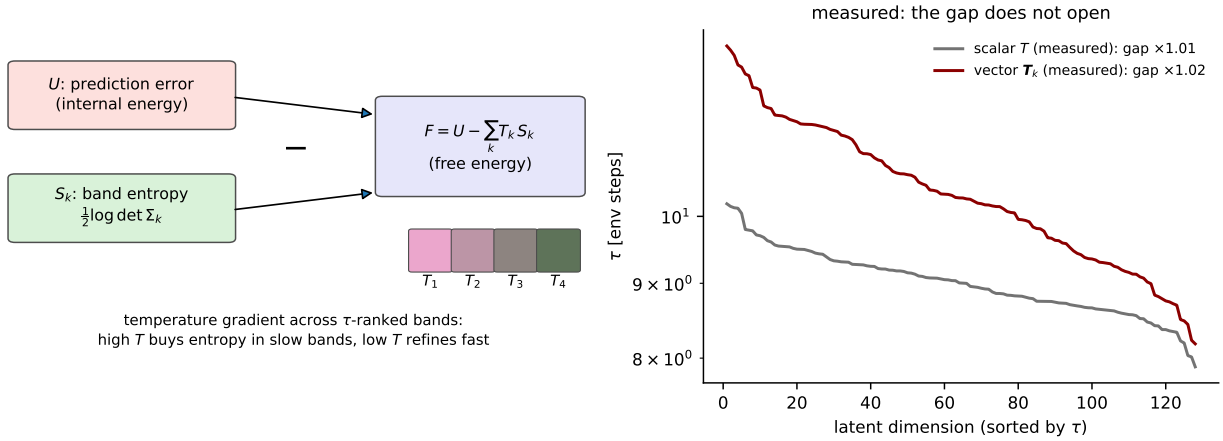


Figure 5: **Mechanism, tested.** Left: the free-energy mapping and the per-band temperature gradient that was trained. Right: the measured spectra of the scalar and vector arms—the vector moves every redistribution metric in the intended direction, but the spectra overlap; the gap does not open.

measurement contribution—the τ referee, the universality of spectral gap collapse, and the two-dead-ends diagnosis—stands independently of which way the exploration resolves.

7. Conclusion

Self-organized latents never separate time scales—five training regimes and an independent deployed model agree, and the temperature-vector arm joins them (Fig. 5). The first-principles fix suggested by the measurements redistributes variance but not content: temperature preserves information, and only placement can organize it. What force can place information by time scale inside a trainable objective is genuinely open; this proposal brings to it a referee that measures, criteria fixed in advance, and an endgame that matters for hierarchical planning.

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